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CHAPTER 2 :- MATRICES

Std : 12th Commerce

Subject :- Mathematics Paper-I

Subtopic :- Properties of adjoint of Matrix, elementary
Transformation , Inverse of matrix

Transpose of Matrice

Properties of transpose of a matrix :-

1. If A and B are two matrices of same order then , $(A + B)^T = A^T + B^T$
2. If A is a matrix and K is constant , then $(KA)^T = K A^T$
3. If A and B are comfortable for the product AB then $(AB)^T = B^T A^T$
4. If A is a symmetric matrix ,then $A^T = A$
5. If A is a skew symmetric matrix ,

Let's Note: A square matrix A can be expressed as the sum of a symmetric and a skew symmetric matrix as follows.

$$A = \frac{1}{2} (A + A^T) + \frac{1}{2} (A - A^T)$$

For example: Let $A = \begin{bmatrix} 4 & -5 & 3 \\ -6 & 2 & 1 \\ 7 & 8 & -9 \end{bmatrix}$, $\therefore A^T = \begin{bmatrix} 4 & -6 & 7 \\ -5 & 2 & 8 \\ 3 & 1 & -9 \end{bmatrix}$

$$A + A^T = \begin{bmatrix} 4 & -5 & 3 \\ -6 & 2 & 1 \\ 7 & 8 & -9 \end{bmatrix} + \begin{bmatrix} 4 & -6 & 7 \\ -5 & 2 & 8 \\ 3 & 1 & -9 \end{bmatrix} = \begin{bmatrix} 8 & -11 & 10 \\ -11 & 4 & 9 \\ 10 & 9 & -18 \end{bmatrix}$$

$$\text{Let } P = \frac{1}{2} (A + A^T) = \frac{1}{2} \begin{bmatrix} 8 & -11 & 10 \\ -11 & 4 & 9 \\ 10 & 9 & -18 \end{bmatrix} = \begin{bmatrix} 4 & -\frac{11}{2} & 5 \\ -\frac{11}{2} & 2 & \frac{9}{2} \\ 5 & \frac{9}{2} & -18 \end{bmatrix}$$

The matrix P is a symmetric matrix.

$$\text{Also } A - A^T = \begin{bmatrix} 4 & -5 & 3 \\ -6 & 2 & 1 \\ 7 & 8 & -9 \end{bmatrix} - \begin{bmatrix} 4 & -6 & 7 \\ -5 & 2 & 8 \\ 3 & 1 & -9 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -4 \\ -1 & 0 & -7 \\ 4 & 7 & 0 \end{bmatrix}$$

$$\text{Let } Q = \frac{1}{2} (A - A^T) = \frac{1}{2} \begin{bmatrix} 0 & 1 & -4 \\ -1 & 0 & -7 \\ 4 & 7 & 0 \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{2} & -2 \\ -\frac{1}{2} & 0 & -\frac{7}{2} \\ 2 & \frac{7}{2} & 0 \end{bmatrix}$$

The matrix Q is a skew symmetric matrix.

Since $P + Q = \text{symmetric matrix} + \text{skew symmetric matrix}$.

Thus $A = P + Q$.

Elementary Transformation :-

There are three elementary transformations of a matrix , They can be used on rows and columns.

1.Interchange of any two rows or any two column :

If we interchange the i^{th} row and j^{th} row of a matrix then , after this interchange , the original matrix is transformed to a new matrix. This transformation is symbolically denoted as $R_i \leftrightarrow R_j$ or R_{ij} .

The same transformation can be applied to two column, say $C_i \leftrightarrow C_j$ or C_{ij}

For Example:

If $A = \begin{bmatrix} 5 & 7 \\ 3 & 2 \end{bmatrix}$ then $R_1 \leftrightarrow R_2$ gives new matrix $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$

and $C_1 \leftrightarrow C_2$ gives new matrix $\begin{bmatrix} 7 & 5 \\ 2 & 3 \end{bmatrix}$

2. Multiplication of the element of any row or column by a non zero scalar :-

If K is a non zero scalar and the row R_i is to be multiplied by a scalar k , then we multiply every element of R_i by the scalar k . symbolically the transformation is denoted by KR_i or $R_i(k)$.

For example: If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 0 \\ 4 & 1 & 3 \end{bmatrix}$ then $4R_2$ gives $\begin{bmatrix} 1 & 2 & 3 \\ 8 & 20 & 0 \\ 4 & 1 & 3 \end{bmatrix}$

Similarly, if any column of the matrix is multiplied by a constant then we multiply every element of the column by the constant. It is denoted by kC_i or $C_i(k)$

If $B = \begin{bmatrix} 2 & 3 \\ -5 & 1 \end{bmatrix}$ then 3C_2 gives $\begin{bmatrix} 2 & 9 \\ -5 & 3 \end{bmatrix}$

3. Adding the scalar multiples of all the elements of any row (column) to corresponding elements of any other row (column) :-

If k is a non Zero scalar and the k -multiples of the elements of R_j (C_j) are added to the elements of R_i (C_i) then the transformation is symbolically denoted as $R_i + kR_j$ or $C_i + kC_j$

For example:

$$\begin{aligned} 1) \quad \text{If } A = \begin{bmatrix} 2 & 5 \\ 7 & 8 \end{bmatrix} \text{ then } R_2 + 2R_1 \text{ gives } &= \begin{bmatrix} 2 & 5 \\ 7+2(2) & 8+2(5) \end{bmatrix} \\ &= \begin{bmatrix} 2 & 5 \\ 11 & 18 \end{bmatrix} \end{aligned}$$

Note :-

- 1) After transformation $R_i + kR_j$, R_j remains same as in the original matrix. Similarly with the transformation $C_i + kC_j$, C_j remains same as in the original matrix.
- 2) The elements of a row or multiples of the element of a row can not be added to the elements of a column or conversely.
- 3) When any elementary row transformations are applied on both the sides of $AB = C$, the prefactor A changes and B remains unchanged. The same row transformations are applied on C .

Inverse of matrix :-

If A is a square matrix of order m and if there exists another square matrix B of the same order such that $AB = BA = I$, where I is the unit matrix of order m then B is called the inverse of A and is denoted by A^{-1} (read as A inverse)

Using the notation A^{-1} for B we write the above equations as $AA^{-1} = A^{-1}A = I$

Note :- For the the existence of inverse of matrix A , it is necessary that $|A| \neq 0$
 A is a non singular matrix .

Uniueness of the inverse of matrix :-

If A is a Square matrix where $|A| \neq 0$ then its inverse ,say A^{-1} is unique.

Theorem :-

Prove that if A is square matrix and its inverse exists then the inverse is unique.

Proof: Let A be a square matrix of order m and let its inverse exist.

Let, if possible, B and C be two inverses of A

Then, by definition of the inverse matrix,

$$AB = BA = I \text{ and } AC = CA = I$$

$$\text{Now consider } B = BI$$

$$= B(AC)$$

$$= (BA)C$$

$$= IC$$

$$B = C$$

Hence $B = C$, that is, the inverse of a matrix is unique.

For exercise 2.4 and 2.5 follow my solved exercise photos .

Thank You

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Reference by Maharashtra state board part- I

Mathematics Text book of Commerce .